Constraints

$\mathcal{X} = \{x \mid A_x x \leq b_x\}$ $\mathcal{U} = \{u \mid A_u u \leq b_u\}$ $\mathcal{X}_f = \{x \mid A_f x \leq b_f\}$

Constrained LQR without Substitution

$J_0^*(x(0)) = \min_{U_0} [z'x(0)'] \begin{pmatrix} H & 0 \\ 0 & Q \end{pmatrix} [z'x(0)]'$

subject to $G_{0,\text{in}} z \leq w_{0,\text{in}} + E_{0,\text{in}} x(0)$

$= G_{0,\text{in}} z$

$= E_{0,\text{in}} x(0)$

Constrained LQR with Substitution

$J_0^*(x(0)) = U_0' H U_0 + 2x(0)' F U_0 + x(0)' Y x(0)$

$= [U_0' x(0)'] \begin{pmatrix} H & F' \\ F & Y \end{pmatrix} [U_0' x(0)']'$

subject to $G_0 U_0 \leq w_0 + E_0 x(0)$

$x(0), U_0^*$ can be found with QP solver

Optimal Input

$U_0^* = K_i x + g_i \quad \text{if } x \in \mathcal{CR}_i \quad \bigcup_i \mathcal{CR}_i = \mathcal{X}_0$

$\mathcal{CR}_i$: polyhedral critical regions

Constrained LQR

$J_0^*(x(0)) = \min_{U_0} J_0(x(0), U_0)$

subject to $x_{k+1} = Ax_k + Bu_k, k = 0, ..., T-1$

$x_k \in \mathcal{X}, u_k \in \mathcal{U}, k = 0, ..., T-1$

$x_T \in \mathcal{X}_f$

$x_0 = x(0)$

Feasible Sets

Set of initial states for which the optimal control is feasible:

$\mathcal{X}_0 = \{x_0 \in \mathbb{R}^n \mid \exists (u_0, ..., u_{T-1}) \text{ such that } x_k \in \mathcal{X}, u_k \in \mathcal{U}, k = 0, ..., T-1, x_T \in \mathcal{X}_f, \text{where } x_{k+1} = Ax_k + Bu_k\}$

Set of states for which the optimal control is feasible

$\mathcal{X}_i = \{x_i \in \mathbb{R}^n \mid \exists (u_i, ..., u_{T-1}) \text{ such that } x_k \in \mathcal{X}, u_k \in \mathcal{U}, k = i, ..., T-1, x_T \in \mathcal{X}_f, \text{where } x_{k+1} = Ax_k + Bu_k\}$

independent of cost

Batch Approach

open-loop solution

Unconstrained LQR

Set the gradient to zero:

$\nabla_{U_0} = J_0(x(0), U_0) = 2HU_0 + 2F'x(0) = 0$

$\Rightarrow U_0^*(x(0)) = -H^{-1}F'x(0)$

$= -\left(S^{x'}\bar{Q}S^x + \bar{R}\right)^{-1} S^{x'}\bar{Q}S^x x(0)$

$= \boldsymbol{K}x(0)$

The resulting optimal cost:

$J_0^*(x(0)) = -x(0)'F'HFx(0) + x(0)'S^{x'}\bar{Q}S^x x(0)$

$= x(0)'\left(S^{x'}\bar{Q}S^x - S^{x'}\bar{Q}S^u\left(S^{u'}\bar{Q}S^u + \bar{R}\right)^{-1} S^{u'}\bar{Q}S^x\right)x(0)$

Quadratic cost function

$J_0(x_0, U_0) = x_T' P x_T + \sum_{k=0}^{N-1} \left[x_k' Q x_k + u_k' R u_k\right]$

$P \geq 0, P' = P$ terminal weight

$Q \geq 0, Q' = Q$ state weight

$R > 0, R' = R$ input weight

STRICTLY CONVEX

Continuous-Time Optimal Control

$J_{0\to T_f}^*(x(0)) = \min_{U_{0\to T_f}} p(x(T_f)) + \int_0^{T_f} q(x(t), u(t))$

s.t. $\dot{x} = f_{(x(t),u(t))}, t \in [0, T_f]$

$h(x(t), u(t)) \leq 0, t \in [0, T_f]$

$x(T_f) \in \mathcal{X}_f$

$x_0 = x(0)$

infinite-dimensional optimization problem

Optimize over input sequences

$\min_{U_{0\to T}} J_{0\to T}(x(0), U_{0\to T})$

where,

$U_{0\to T} = [u_0, u_1, ..., u_{T-1}]$

Batch Approach

Optimal u

Implement

$u(k) = u_k^*$

Real system or simulator

$x(k+1) = f_{\text{real}}(x(k), u(k))$

Prediction Model

$x_{k+1} = f(x_k, u_k)$

Task horizon T

Number of time steps in the task

Cost Function $J_{0\to T}$

stage cost q summed over the horizon plus terminal cost p

CFTOC

Constrained Finite Time Optimal Control

$\mathcal{X}_{0\to T} \subseteq \mathbb{R}^n$ is the set of feasible initial conditions

$J_{0\to T}^*(x(0))$ is the value function (optimal cost)

$U_{0\to T}^* = [u_0^*, ..., u_{T-1}^*]$ is one of the minima

Finite horizon optimal control

$J_{0\to T}^*(x(0)) = \min_{U_{0\to T}} J_{0\to T}(x(0), U_{0\to T})$

subject to $x_{k+1} = f(x_k, u_k), k = 0, 1, ..., T-1$

$h(x_k, u_k) \leq 0, k = 0, 1, ..., T-1$

$x_T \in \mathcal{X}_f$

$x_0 = x(0)$

Optimal Control

$\min_{x_0, ..., x_{T-1}, u_0, ..., u_{T-1}} \sum_{k=0}^{T-1} q(x_k, u_k) + p(x_T)$

s.t. $x_{k+1} = f(x_k, u_k)$

$u_k = \pi_k(x_k)$

$x_k \in \mathcal{X}, u_k \in \mathcal{U}$

$x_T \in \mathcal{X}_f$

$x_0 = x(0)$

Control Policy

$u_k = \pi_k(x_k)$

state and input constraints

$x_k \in \mathcal{X}, u_k \in \mathcal{U}$

Terminal constraint

$x_T \in \mathcal{X}_f$

Modeling

Experiential Control Design

Optimal value (infimum)

$J^* = J^* = p^* = f(z^*)$

lowest possible objective value

Strictly feasible point

A vector $z \in \mathcal{Z}$ satisfying the inequality constraints strictly that is

$g_{i(z)} < 0 \quad \forall i = 1, ..., m$

Rewriting max to min

$\max f(z) \to \min -f(z)$

and, solve for

$z_{\max}^* \to z_{\min}^*$

Optimal soution (minimizer)

Any feasible $z^* \in \mathcal{Z}$ s.t. $f(z^*) \leq f(z)$ for all feasible $z \in \mathcal{Z}$

Feasible point

A vector $z \in \mathcal{Z}$ satisfying the inequality and equality constrains

General Formulation

$\min f(z)$

subject to $z \in S \subseteq \mathcal{Z}$

z collects decision variables

$\mathcal{Z}$ domain of decision variables

$S \subseteq \mathcal{Z}$ is the constraint set with feasible decisions

$f : \mathcal{Z} \to \mathbb{R}$ assigns a cost f to each decision z

Optimizations

Least possible cost

$f(z^*) = f^*$

Local optimality

z^* is locally optimal if there exists and $R > 0$ s.t. $z = z^*$ is optimal for $\|z - z^*\|_2 \leq R$

Terminology

Redundant

Redundant if the constraint does not effect the feasible set.

Removing does not change the solution.

Active/Inactive constraint

Active if the inequality constraint if $g_i(\bar{z}) = 0$

Else the constraint is inactive.

Explicit Constraint

g and h are explicit constraints.

Implicit

Constraint defined by the function to be minimized

e.g. $\log(z) \to z \in \mathbb{R}^+$

Classical unconstrained

e.g. min of a linear function

$c^\top z + k$

or min of a positive definite quadratic function

$z^\top H z + c^\top z + k$

Definitions

Positive Definite

Symmetric and $H \geq 0$

iff $\text{eig}(H) \geq 0$

Negative Definite

Symmetric and $H \leq 0$

Indefinite

Unbounded problem

$z^* = \text{DNE}$

Norm

$\|z\|_2 = 2\text{-norm of } z = \sqrt{z_1^2 + z_2^2 + ... + z_n^2}$

$\|z\|_1 = 1\text{-norm of } z = |z_1^2| + |z_2^2| + ... + |z_n^2|$

$\|z\|_\infty = \infty\text{-norm of } z = \max\left\{|z_1^2| + |z_2^2| + ... + |z_n^2|\right\}$

Epigraph

$\text{epi}(f) = \left\{\begin{pmatrix} z \\ t \end{pmatrix} \;\middle|\; z \in \text{dom}(f), f(z) \leq t\right\} \subseteq \text{dom}(f) \times \mathbb{R}$

a function is convex iff its epigraph is a convex set.

it has dimension one higher than the domain of f.

Nonlinear Programming (NLP) Formulation

$\min_z f(z)$

subject to $g_i(z) \leq 0 \quad i = 1, ..., m$

$h_j(z) = 0 \quad j = 1, ..., p$

Domain $\mathcal{Z} \subseteq \mathbb{R}^n$ of the objective function, from which decision variabe is taken

Optional inequality constraint functions $g_i : \mathbb{R}^n \to \mathbb{R}$

Optional inequality constraint functions $h_j : \mathbb{R}^n \to \mathbb{R}$

Properties

$J^* = \min_{z \in \mathcal{Z}} f(z^*)$

If $J^* = -\infty \to$ unbounded below

If S is empty then the problem is infeasible ($J^* := -\infty$)

If $S = \mathcal{Z} \to$ unconstrained

Linear Programming (LP) formulation

$\min_z c^\top z$

subject to $Gz \leq h$

$Az = b$

Linear cost, linear constraints, feasible set is polyhedron

Quadratic Program (QP)

$\min_z \frac{1}{2} z^\top P z + q^\top z$

subject to $Gz \leq h$

$Az = b$

Convex if $P \geq 0$

Nonconvex if $P \not\geq 0$

Quadratic cost and linear constraints produces polyhedron feasible set

General Quadratic Program

$\min_z \frac{1}{2} z^\top P z + q^\top z + r, \quad P \geq 0$

subject to $Gz \leq h$

$Az = b$

with $P \in \mathbb{S}_+^n$

General linear Program

Affine cost and constraints:

$\min c^\top z + d$

subject to $Gz \leq h$

$Az = b$

Constant d can be left out, it has no effect on optimal solution

Piecewise Affine Minimization

$\min_z \left(\max_{i=1,...,m} \left(c_i^\top z + d_i\right)\right)$

subject to $Gz \leq h$

Infinity Norm/Chebyshev Minimization

$\min_z \|z\|_\infty$

subject to $Fz \leq g$

Equivalent to:

$\min_{z,t} \{\max\{z_1, ..., z_n, -z_1, ..., -z_n\}\}$

subject to $Fz \leq g$

adding variables

$\min_{z,t} t$

subject to $z_i \leq t, \quad i = 1, ..., n$

$-z_i \leq t, \quad i = 1, ..., n$

$Fz \leq g$

notating with a vector of ones

$\min_{z,t} t$

subject to $\mathbf{1}z \leq t\mathbf{1}$

$Fz \leq g$

Equivalent LP for piecewise

$\min_{z,t} t$

subject to $c_i^\top z + d_i \leq t, \quad i = 1, ..., m$

$Gz \leq h$

added decision variables and problem written in epigraph form

One Norm Minimization

$\min_z \|Az - b\|_1$

subject to $Fz \leq g$

re-written as a max of linear functions:

with $A \in \mathbb{R}^{m \times n}$

$\min_{z,t} \sum_{i=1}^m \max\{(Az - b)_i, -(Az - b)_i\}$

subject to $Fz \leq g$

adding variables

$\min_{z \in \mathbb{R}^n, t \in \mathbb{R}^m} t_1 + \cdots + t_m$

subject to $(Az - b)_i \leq t_i, \quad i = 1, ..., n$

$-(Az - b)_i \leq t_i, \quad i = 1, ..., n$

$Fz \leq g$

notating with a vector of ones

$\min_{z,t} \mathbf{1}^\top t$

subject to $-t \leq (Az - b) \leq t$

$Fz \leq g$

Alternative format

$\min_z c^\top z$

subject to $Az = b$

$z \geq 0$

Can convert previous format with extra variables

Convex Functions

A function $f : \text{dom}(f) \to \mathbb{R}$ is convex iff its domain $\text{dom}(f)$ is convex and

$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y), \quad \forall \lambda \in (0,1), \quad \forall x, y \in \text{dom}(f)$

Examples:

Affine: $ax + b$

Exponential: e^{ax}

Powers: x^α on domain $\mathbb{R}_+$ for $\alpha \geq 1 \vee \alpha \leq 0$

Powers of absolute value: $|x|^p$ for $p \geq 1$

Convexity preserving operations

Non-negative weighted sum $\alpha f, \alpha \geq 0$

Composition with affine function $f(Ax + b)$

Pointwise max $f(z) = \max\{f_1(z), ..., f_m(z)\}$

Convex Optimization

f is a convex function over domain $\mathcal{Z}$

S is a convex set

1st order condition

$f(y) \geq f(z) + \nabla f(z)^\top (y - z), \quad \forall z, y \in \text{dom}(f)$

1st order approximator of f around any point z is a global underestimator of f

2nd order condition

$\nabla^2 f(z) \succeq 0, \quad \forall z \in \text{dom}(f)$

Strictly convex

if the inequality is strict

Level set

$L_\alpha = \{z \mid z \in \text{dom}(f), \quad f(z) = \alpha\}$

set of all points where a given function has a constant value

Sublevel set

$C_\alpha = \{z \mid z \in \text{dom}(f), \quad f(z) \leq \alpha\}$

if f is convex, the sublevel sets of f are convex for all α

Concave

f is concave iff the function $-f$ is convex

Examples:

Affine: $ax + b$

Powers: x^α on domain $\mathbb{R}_+$ for $0 \leq \alpha \leq 1$

Log: $\log(x)$ on domain $\mathbb{R}_+$

Entropy: $-x \log(x)$ on domain $\mathbb{R}_+$

Convex problem

$\min_{z \in \mathcal{Z}} f(z)$

subject to $g_i(z) \leq 0 \quad$ (convex functions) $\quad i = 1, ..., m$

$a_j^\top z = b_j \quad j = 1, ..., p$

Affine sets

$Z = \{z \in \mathbb{R}^n \mid Az = b\}$

a subspace is an affine set where, $b = 0$

l_p norm

Denoted as $\|z\|_p$ on $\mathbb{R}^n$ and defined for any $p \geq 1$ by

$\|z\|_p := \left(\sum_{i=1}^n |z_i|^p\right)^{\frac{1}{p}}$

Vector Norm

Any function $f : \mathbb{R}^n \to \mathbb{R}$ satisfying:

1. $f(x) \geq 0 \wedge f(z) = 0 \Rightarrow z = 0$
2. $f(tx) = |t| f(z)$ for scalar t
3. $f(z + y) \leq f(z) + f(y), \forall z, y \in \mathbb{R}^n$

Denoted as $\|z\|$

Norm ball

$\{z \mid \|z - z_c\| \leq r\}$

always convex for any norm

Convex sets

$\mathcal{Z}$ is convex $\Leftrightarrow \lambda z + (1-\lambda)y \in \mathcal{Z}, \forall \lambda \in [0,1], \forall z, y \in \mathcal{Z}$

set $\mathcal{Z}$ is convex iff for any pair of points z and y in $\mathcal{Z}$, any convex combination of z and y lie in $\mathcal{Z}$

Hyperplanes

$\mathcal{Z} = \{z \in \mathbb{R}^n \mid a^\top z = b\}$ for $a \neq 0$, where $a \in \mathbb{R}^n$

is the normal vector to the hyperplane

Halfspaces

$\mathcal{Z} = \{z \in \mathbb{R}^n \mid a^\top z \leq b\}$

open or closed, everything one one side of a hyperplane

Unions

The union of two or more convex sets is not convex in general

Intersections

Intersection of two or more convex sets is convex

Convex Hull

Smallest convex set that contains $\mathcal{Z}$

$\text{Co}(Z) := \{z \mid z\lambda a + (1-\lambda)b, \quad \lambda \in [0,1], \quad a, b \in Z\}$

basically the set of halfspaces bounding a set of points. For a set $\mathcal{Z} = \{z_1, z_2, ..., z_q\}$ the convex hull is:

$\text{Co}(Z) = \left\{\lambda_1 z_1 + \lambda_2 z_2 + \cdots + \lambda_q z_q \mid z_\lambda \geq 0, \sum_{i=1}^q \lambda_i = 1\right\}$

Polyhedron

$Z = \{z \mid a_1^\top z \leq b_1, a_2^\top z \leq b_2, ..., a_m^\top m \leq b_m\}$

$= \{z \mid Az \leq b\}$

the intersection of a finite number of closed halfspaces

Polytope

A bounded polyhedron.

the intersections occur such that a closed polyhedron is formed

Optimality Criterion for Differentiable f

Theorem (Necessary Condition*)

$f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at z^*.

If z^* is a local minimizer then $\nabla f(z^*) = 0$

Theorem (Sufficient conditions*)

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is twice differentiable at z^*.

If $\nabla f(z^*) = 0$ and the Hessian of $f(z)$ at z^* is positive definite, then z^* is a local minimizer.

Theorem (Necessary and sufficient condition*)

Suppose that $f : \mathbb{R}^n \to \mathbb{R} = z^\top H z + c^\top z + k$. If H is positive definite, then z^* is the global minimizer iff $\nabla f(z^*) = 0$.

Unconstrained optimization using gradients

$z^{k+1} = z^k - h^k \nabla f(z^k)$

h^k is the chosen step size.

Local/Global Optimality Proof

Assume convex problem and let z^* be a local optimal solution, then z^* is a global optimal solution.

Optimality Conditions

1. Unconstrained with $x > 1 \to$ use gradient method
2. Constraint $\to$ use KKTs

Steepest descent

$\boldsymbol{d}_x = -\frac{\nabla f(\bar{z})}{|\nabla f(\bar{z})|}$

$\boldsymbol{d}_x$ is the direction of steepest descent that minimizes $\nabla f(\bar{z})' \boldsymbol{d}_x < 0$

Descent Direction for Differentiable f

Theorem (Descent Direction)

$f : \mathbb{R}^n \to \mathbb{R}$ differentiable at $\bar{z}$. If there exists a vector $\boldsymbol{d}$ s.t. $\nabla f(\bar{z})^\top \boldsymbol{d} < 0$, then there exists a $\delta > 0$ s.t $f(\bar{z} + \lambda \boldsymbol{d}) < f(\bar{z}) \quad \forall \lambda \in (0, \delta)$.

$\boldsymbol{d}$ is the descent direction.

KKT Optimality Conditions

Karush-Kuhn-Tucker

Dual-Variables (u^*, v^*) = Lagrange Multipliers

1. Stationarity conditions: $0 = \nabla f_{z^*} + \sum_{i=1}^m u_i^* \nabla g_i(z^*) + \sum_{j=1}^p v_j^* \nabla h_j(z^*)$
2. Complementary Slackness conditions: $0 = u_i^* g_i^*, \quad i = 1, ..., m$
3. Dual Feasibility: $0 \leq u_i^*, \quad i = 1, ..., m$
4. Primal Feasibility: $\begin{cases} g(z_{\text{opt}})^{(z^*)}, & i=1...m \\ h(z_{\text{opt}})^{(z^*)}, & j=1...p \end{cases}$

Theorem (Necessary and Sufficient condition)

Suppose that the NLP is convex and that cost and constraints f, g_i and h_j are differentiable at a feasible z^*. If the NLP satisfies constraint qualification conditions then z^* is optimal iff there are (u^*, v^*) that, together with z^*, satisfy the KKT conditions.

Duality Theory

Any feasible point: upper bound of the optimal value

Lagrange dual problem: lower bound on optimal value

The Lagrange Function

$L(z, u, v) = f(z) + u_1 g_1(z) + ... + u_m g_m(z) + v_1 h_1(z) + \cdots + v_p h_p(z)$

$= f(z) + u^\top g(z) + v^\top h(z)$

objective plus weighted sum of constraints

For feasible z and arbitrary vectors $u \geq 0$ and v, we have:

$L \leq f$

After minimization

$\min_{z \in \mathcal{Z}} L \leq \min_{z \in \mathcal{Z}, g(z) \leq 0, h(z) = 0} f(z) = p^*$

Dual Function

$d(u, v) \triangleq \min_{z \in \mathcal{Z}} L(z, u, v) \in [-\infty, \infty]$

Computing $d(u, v)$ is the dual subproblem.

This is an unconstrained optimization problem.

$d(u, v)$ is always concave.

The point (u, v) is dual feasible if $u \geq 0$ and $(u, v) \in \text{dom}(d)$.

$d(u, v)$ is always a lower bound for p^* for any feasible (u, v).

Lagrangian Dual Problem

Best choice for lower bound

$\max_{(u,v): u \geq 0} d(u, v)$

The dual problem is always convex.

(u, v) is dual feasible if $u \geq 0$ and $(u, v) \in \text{dom}(d)$.

Weak Duality

$\max_{(u,v): \quad u \geq 0} d(u, v) \leq \min_{z \in \mathcal{Z}, g(z) \leq 0, h(z) = 0} f(z)$

Duality Gap and Certificate of Optimality

$d^* = \max_{(u,v): \quad u \geq 0} d(u, v)$

$p^* = \min_{z \in \mathcal{Z}, g(z) \leq 0, h(z) = 0} f(z)$

We always have $d^* \leq p^*$

$|d^* - p^*|$ is the optimal duality gap

Strong duality $d^* = p^* \to u^*$ and v^* serve as certificate of optimality

Mixed Integer Linear Programs (MILP)

$\min_z c^\top z$

subject to $Gz \leq h$

$Az = b$

$z \in \{0, 1\}^n$ or $z \in \mathbb{Z}^n$

linear program with some integer or binary constraints

Lifting

Additional decision variable to make "easer" problem

$\min_{z,t} |z| \to \min_{z,t,\varepsilon} \varepsilon$

and update the constraints accordingly

Least squares

$\min_z \|Az - b\|_2^2$

unconstrained solution is $A^\dagger b$

Relaxation

relaxed constraint becomes

$0 \leq z_i \leq 1 \quad i = 1, ..., s$